

Plasma Column Module — Axial Injection

Purpose: Beam-ionized (self-neutralizing) plasma column inside an axial solenoid to mitigate space-charge blow-up between the last solenoid and the spiral/electrostatic inflector of a compact cyclotron.

Operating Point

- **Beam:** proton (p^+), **30 keV, 10 mA DC** (or long macropulse)
- **Nominal beam radius:** 1–5 mm at column entrance
- **Residual gas:** H_2 baseline (Kr optional for faster neutralization)
- **Nominal local pressure (cell):** 1×10^{-5} torr (with differential pumping to protect cyclotron tank)
- **Magnetic field (axial):** 0.10–0.20 T uniform over active length

Neutralization (order-of-magnitude)

- H_2 @ 1×10^{-5} torr $\rightarrow \tau \approx 0.06$ ms to reach $\sim n_e \approx n_b$
- Kr @ 1×10^{-5} torr $\rightarrow \tau \approx 0.013$ ms ($\approx 5 \times$ faster)
- Electron gyroradius at 1 eV: $\sim 17\text{--}34$ μm for 0.2–0.1 T $\rightarrow$ strongly magnetized

This is a **neutralizing transport** section (space-charge compensation, mild ion-focusing), **not** a current-carrying plasma lens.

Mechanical & Magnetic

- **Overall length (flange-to-flange):** 260 mm (fits 100–300 mm active column variants)
- **Active column length L:** configurable 100 / 200 / 300 mm (spacer rings)
- **Clear beam aperture (ID):** ≥ 16 mm (recommend $\varnothing 20$ mm beam tube, smooth, non-magnetic SS 316L or Al 6061-T6, $R_a \leq 0.8$ μm)
- **Solenoid:** water-cooled, $B_z = 0.10\text{--}0.20$ T, effective magnetic length $\approx L$, bore $\varnothing \geq 40$ mm

- **Uniformity:** $\Delta B/B \leq \pm 2\%$ over $\varnothing 10 \text{ mm} \times L$
 - **Fringe-field trim:** auxiliary Helmholtz/trim coils ($\pm 5\%$ correction)
 - **Alignment:** beamline fiducials, mechanical axis to beam axis $\leq \pm 0.2 \text{ mm}$, tilt $\leq 0.3 \text{ mrad}$
-

Vacuum & Differential Pumping

- **Cell pressure setpoint:** $1 \times 10^{-5} \text{ torr}$ (H_2); optional **Kr seeding** up to $1 \times 10^{-5} \text{ torr}$ via MFC
 - **Upstream isolation:** $\varnothing 6\text{--}8 \text{ mm}$ orifice + **TMP** $\geq 500 \text{ L/s}$ (H_2 equivalent)
 - **Downstream isolation (toward inflector):** $\varnothing 4\text{--}6 \text{ mm}$ orifice + **TMP** $\geq 700 \text{ L/s}$ + cryopanel (if available)
 - **Cyclotron tank target:** $\leq 1 \times 10^{-7} \text{ torr}$ during operation
 - **Materials:** all-metal UHV seals (CF), vented screws, low-outgassing cables; **no organics** in beam path
-

Electrodes & Biasing (Electron Containment)

- **End electrodes:** two axial "cups"/grids, $\varnothing \geq 18 \text{ mm}$ clear, edge-rounded; mounted flush with column ends
 - **Bias:** **+5 to +30 V** w.r.t. beam pipe (optimize in situ)
 - **Purpose:** set plasma potential, reduce axial e^- loss, speed up neutralization
 - **Guard sleeves:** floating/grounded sleeves along the inner wall to suppress secondary electrons; optional **+5 V** bias
 - **HV compatibility:** insulation to 200 V, $< 10 \text{ nA}$ leakage at vacuum
-

Diagnostics & Controls

- **Ports:** 2× mini-ConFlat viewports (plasma light), 2× Langmuir-pin feedthroughs (optional), 1× B-dot/flux probe feedthrough

- **Beam:** retractable **scintillator/OTR screen** at exit ($\varnothing$ 20 mm), **DCCT/Faraday cup** upstream
 - **Pressure:** 2× nude Bayard–Alpert gauges (cell & downstream), 1× RGA port (shared manifold)
 - **Interlocks:** close Kr MFC and ramp solenoid to **<0.05 T** on cyclotron vacuum trip; inhibit beam if cell pressure **> 5×10^{-5} torr**
-

Electrical & Thermal

- **Solenoid PS:** 0–200 A, <10 mA ripple (50/60 Hz); field stability $\leq 0.1\%$ over minutes
 - **Cooling:** 1–2 L/min DI water, $\Delta T < 10^\circ\text{C}$; bakeable to 150°C (solenoid removed)
 - **Electrodes PS:** 0–50 V, 100 mA max, floating or ground-referenced, bandwidth ≥ 1 kHz
-

Expected Performance (30 keV, 10 mA p^+)

- **Neutralization fraction:** >90% steady-state at 1×10^{-5} torr (H_2); faster onset with Kr
 - **Emittance growth:** reduced vs vacuum-only transport; minimal additional scattering over 0.1–0.3 m at 10^{-5} torr
 - **Matching:** choose **$L = 0.2$ m** and **$B_z \approx 0.15\text{--}0.20$ T** to minimize divergence into inflector; tune end-electrode bias for max transmission
-

Integration Notes

- Place module as **close as mechanically possible** to the inflector entrance while keeping the **downstream orifice** ahead of the inflector to shield tank vacuum
 - Provide **straight-through line of sight** for alignment laser to median plane fiducials
 - Commissioning sequence: pump-down → solenoid at 0.15 T → beam at low current → set biases (+10 V start) → raise to 10 mA → optimize Larmor confinement & pressure
-

Optional Krypton (Kr) Seeding — “Fast-Start” Plasma Column Mode

Goal: accelerate neutralization build-up and improve stability by adding a small **Kr partial pressure** within the module while protecting the cyclotron tank with differential pumping.

Operating Envelope

- **Kr partial pressure (cell):** $(3\text{--}10)\times 10^{-6}$ torr (with H_2 making up the balance to $\sim 1\times 10^{-5}$ torr total), or **Kr-only** at 1×10^{-5} torr during commissioning tests.
- **Expected neutralization time (30 keV p^+):** with $\sigma_{\text{Kr}} \approx 1\times 10^{-19} \text{ m}^2 \rightarrow \tau \approx 0.013 \text{ ms}$ at 1×10^{-5} torr ($\approx 5\times$ faster than H_2 at same pressure).
- **Recommended active length:** $L = 0.2 \text{ m}$; $B_z = 0.15\text{--}0.20 \text{ T}$ for strong electron magnetization.

Gas Handling & Hardware

- **Injection location:** mid-cell side port, facing upstream to minimize gas line-of-sight into the inflector region.
- **Flow control:** MFC (0–1 sccm Kr) + **fast piezo valve** for pulsed puffs (1–50 ms), both behind a **conductance-limiting orifice** ($\varnothing$ 0.2–0.5 mm).
- **Valving:** all-metal right-angle valve at the module inlet; **sector gate valve** downstream toward the cyclotron tank for isolation.
- **Sensing:** add **second nude gauge** dedicated to Kr read-back (cal-factor adjusted), and use RGA for partial-pressure confirmation during setup.

Vacuum & Differential Pumping Adjustments

- Keep **downstream orifice** small ($\varnothing$ 4–6 mm) and pair with **TMP $\geq 700 \text{ L/s}$** on the downstream stage; optional **LN₂ cryopanel** if available.
- Interlock **Kr MFC/valve** closed if: tank pressure $> 1\times 10^{-7}$ torr, cell pressure $> 5\times 10^{-5}$ torr, or inflector HV trips.
- For steady operation, aim **Kr-only** or **Kr-dominant** during the **first 100–300 ms** of beam, then taper to **H₂-dominant** to reduce scattering and pump load.

Electrode Bias with Kr

- Start with **+10 V** end-cup bias; scan **+5 → +30 V** to minimize electron end losses. Kr typically yields a slightly **higher plasma potential**, so optimum may shift **+2–5 V** vs H₂.

Beam-Physics Notes (30 keV, 10 mA p⁺)

- **Faster neutralization:** Kr's larger ionization cross-section yields rapid build-up; beneficial for macropulsed operation or quick turn-on.
- **Scattering & loss:** Heavier Kr increases small-angle scattering and energy-loss per path length vs H₂. At $\leq 1 \times 10^{-5}$ torr over **0.1–0.3 m**, this remains **small**, but keep Kr as **low as needed** to meet transmission goals.
- **Tuning strategy:** Commission with **Kr-only @ 1×10^{-5} torr** to establish electrode bias and B-field setpoints; then reduce Kr (or switch to H₂-only) and verify that transmission/emittance remain within spec.

Controls & Sequencing (example)

1. Pump-down to base ($< 1 \times 10^{-7}$ torr tank; $< 5 \times 10^{-7}$ torr cell).
2. Ramp solenoid to **0.15–0.20 T**.
3. Enable Kr **pulsed puff** (10–50 ms) just before beam; maintain $(3\text{--}10) \times 10^{-6}$ torr during first **100–300 ms** of beam.
4. Set end-cup **+10–20 V**; optimize for minimum beam size on exit screen.
5. Taper Kr (hold H₂ at $\sim \text{few} \times 10^{-6}$ torr or off); confirm tank $\leq 1 \times 10^{-7}$ torr and stable inflector HV.
6. Record setpoints; enable interlocks for auto-revert to H₂-only on any trip.

Safety & Materials

- Kr cylinder with **two-stage regulator**, **VCR-type** stainless lines, **baked/vented** to UHV standards.
- No polymers inside vacuum; shield viewports from direct beam/UV.
- Ensure **oxygen deficiency hazard (ODH)** assessment if venting to enclosed spaces (Kr is inert but heavy).